

Proposal Requirements

Draft Proposal Guidelines

You should use the following outline for your proposal, which shouldn't need more than about **one page**. You should address each of the points below, with a few sentences each:

- Provide an introduction/overview that speaks to the goals of the project, as well as any challenges or limitations you foresee in the approach and/or dataset.
- Find the closest example you can of one existing project or study that is most similar to what you're proposing: does it use the same dataset or address the same prediction task? Include the link, a few sentences of description, and summarize how what you're proposing is different or improved compared to that existing work. This includes the case where the proposed project is a continuation of a previous milestone or course project for which you received credit: you must state that in the proposal and summarize old vs new work. You cannot propose to reuse (i.e. get credit twice for) a previous project's supervised or unsupervised learning work to satisfy the Milestone 2 project requirements. When in doubt, please ask the instructors.
- Part A (Supervised learning)
 - What is the dataset you propose to use? Provide a link to the dataset.
 - Specify the learning approaches and feature representations that are appropriate to this problem: what do you plan to try, and why? (Your choices could change later.)
 - Are there external datasets or tools that you might incorporate to help with the problem? If so, what are they and what additional work is required to prepare/use them?
 - Describe the evaluation and visualization methods you plan to use.
- Part B (Unsupervised learning)
 - What is the dataset you propose to use? Provide a link to the dataset.
 - What are the question(s) about the dataset's structure you want to answer, or goal to achieve?
 - What data manipulation will be necessary for this dataset to prepare it?
 - Specify unsupervised learning approaches and feature representations that are appropriate for this problem.
 - Are there external datasets or tools that you might incorporate to help with the problem?



- How will you evaluate the quality of your results?
- Describe visualizations that would be appropriate as part of evaluating the effectiveness of your methods or characterizing the structure in the dataset.
- Team Planning:
 - Indicate the specific contributions that each team member will make to the project.
 - Include a rough timeline.

Your proposals will be reviewed by two peers from the class and you will take those into consideration when you revise your proposal for review by the instructional team. You will also discuss your proposal directly with your project coach. Your draft proposal should be a Google Doc that you share. Please note that you must enable commenting on your Google Doc to receive peer reviews.

Peer Review Guidelines

You will receive credit for completing reviews of your peers' proposals (see syllabus for grading). Note that your proposal grade will not be based on the content of your peers' reviews. The purpose of the peer reviews is threefold: (1) to gain experience reviewing proposals, (2) to learn about other work going on in class, and (3) to get feedback on how to make your project better. Your reviews should be 5-10 sentences long and should take into consideration the following points:

- **professional:** what would a co-worker think about your review?
- **pleasant:** courtesy goes a long way.
- **helpful:** what sort of advice would you want?
- **scientific:** focus on facts, not opinions.
- **realistic:** keep scope in mind.
- **empathetic:** how would you feel if you received the review you wrote?
- **organized:** make it easy for the recipient to follow your train of thought.

In machine learning, both positive and negative examples are important: the same is true for peer reviews! Thus, it's important to highlight positive things in peer reviews, as well as areas for improvement. Typically you should point to at least two positive things that the authors did well and identify one area where they might spend some time improving their work (always making specific, constructive suggestions about how to do so).

Peer reviews will take the form of comments on the Google Doc containing the proposal and will be coordinated via a Google Sheet.

Per the instructions in the Canvas assignment description, you are expected to complete two (2) peer reviews.

Final Proposal Guidelines

Taking into consideration the feedback from your peers' reviews of your draft proposal, you will revise your proposal and submit it to the course instructors for final proposal review. The course instructors may make suggestions to your plans and will give the official "go-ahead" to proceed with your project if the final proposal is approved. (Otherwise, they will work with you to find an alternative or acceptable variant.)

You will submit the proposal as a pdf document and github repo link via Course. You will also provide a link to the Google Doc version of your report to your project coach.